DRUG INTERACTION RISK DISCOVERY — RELATIONAL SCHEMA

PostgreSQL tables • PK underlined • FK arrows show references • constraints summarized below

DRUGS
PK drug_id : SERIAL
canonical_name : VARCHAR(255) NOT NULL UNIQUE
pubchem_cid : VARCHAR(50)
inchikey : VARCHAR(50)
smiles : TEXT
drug_type : VARCHAR(100)
therapeutic_class : TEXT
created_at : TIMESTAMPTZ NOT NULL

DRUG_SOURCE_MAPPINGS
PK mapping_id : SERIAL
FK drug_id : INT → drugs.drug_id
source_name : VARCHAR(50) NOT NULL
source_drug_id : VARCHAR(50) NOT NULL
source_drug_name : VARCHAR(255) NOT NULL
match_method : VARCHAR(50) NOT NULL
created_at : TIMESTAMPTZ NOT NULL

FK drug_id
FK drug_b_id

DDI_PAIRS
PK pair_id : BIGSERIAL
FK drug_a_id : INT → drugs.drug_id
FK drug_b_id : INT → drugs.drug_id
created_at : TIMESTAMPTZ NOT NULL

DDI_SEVERITIES
PK severity_id : BIGSERIAL
FK pair_id : BIGINT → ddi_pairs.pair_id
severity_level : VARCHAR(20) NOT NULL
source_dataset : VARCHAR(50) NOT NULL DEFAULT DDInter
source_atc_codes : VARCHAR(100)
created_at : TIMESTAMPTZ NOT NULL

FK pair_id
FK pair_id

DDI_MECHANISMS
PK mechanism_id : BIGSERIAL
FK pair_id : BIGINT → ddi_pairs.pair_id
mechanism_category : VARCHAR(100) NOT NULL
source_dataset : VARCHAR(50) NOT NULL DEFAULT MecDDI
created_at : TIMESTAMPTZ NOT NULL

PATIENTS
PK patient_id : SERIAL
mrn : VARCHAR(50) NOT NULL UNIQUE
full_name : VARCHAR(255) NOT NULL
gender : VARCHAR(20) NOT NULL
date_of_birth : DATE NOT NULL
primary_condition : VARCHAR(255)
is_synthetic : BOOLEAN NOT NULL DEFAULT TRUE
created_at : TIMESTAMPTZ NOT NULL

FK drug_id

PRESCRIPTIONS
PK prescription_id : SERIAL
FK patient_id : INT → patients.patient_id
FK drug_id : INT → drugs.drug_id
dosage : VARCHAR(100) NOT NULL
frequency : VARCHAR(100) NOT NULL
status : VARCHAR(20) NOT NULL DEFAULT ACTIVE
prescribed_by : VARCHAR(100) NOT NULL
start_date : DATE NOT NULL
end_date : DATE NULL
created_at : TIMESTAMPTZ NOT NULL

FK patient_id
FK patient_id
FK drug_b_id

RISK_ASSESSMENTS
PK assessment_id : BIGSERIAL
FK patient_id : INT → patients.patient_id
performed_by : VARCHAR(100) NOT NULL (no FK)
max_hop_depth : INT NOT NULL DEFAULT 2
total_active_medications : INT NOT NULL
direct_interaction_count : INT NOT NULL DEFAULT 0
multi_hop_candidate_count : INT NOT NULL DEFAULT 0
assessment_notes : TEXT
created_at : TIMESTAMPTZ NOT NULL

FK assessment_id

ASSESSMENT_FINDINGS
PK finding_id : BIGSERIAL
FK assessment_id : BIGINT → risk_assessments.assessment_id
FK drug_a_id : INT → drugs.drug_id
FK drug_b_id : INT → drugs.drug_id
hop_count : INT NOT NULL
traversal_path : TEXT NOT NULL
severity_level : VARCHAR(20) snapshot
mechanism_category : VARCHAR(100) snapshot
evidence_sources : VARCHAR(100) NOT NULL
finding_type : VARCHAR(50) NOT NULL
created_at : TIMESTAMPTZ NOT NULL

USER_ACCOUNTS
PK user_id : SERIAL
username : VARCHAR(100) NOT NULL UNIQUE
password_hash : VARCHAR(255) NOT NULL
full_name : VARCHAR(255) NOT NULL
role : VARCHAR(20) NOT NULL
is_active : BOOLEAN NOT NULL DEFAULT TRUE
created_at : TIMESTAMPTZ NOT NULL

Key and integrity constraints
• drugs: UNIQUE(canonical_name)
• drug_source_mappings: UNIQUE(source_name, source_drug_id); source_name ∈ {DDInter, MecDDI}
• ddi_pairs: UNIQUE(drug_a_id, drug_b_id); CHECK drug_a_id < drug_b_id
• ddi_severities: UNIQUE(pair_id, source_dataset); severity ∈ {Major, Moderate, Minor, Unknown}
• ddi_mechanisms: UNIQUE(pair_id, mechanism_category, source_dataset)

Patient and transaction constraints
• patients: UNIQUE(mrn); gender ∈ {Male, Female, Other}; is_synthetic = TRUE
• prescriptions: UNIQUE(patient_id, drug_id, start_date); status ∈ {ACTIVE, DISCONTINUED, COMPLETED}; end_date ≥ start_date if present
• risk_assessments: max_hop_depth BETWEEN 1 AND 3
• assessment_findings: hop_count BETWEEN 1 AND 3; finding_type ∈ {DOCUMENTED_DIRECT, CANDIDATE_PATHWAY}
• user_accounts: UNIQUE(username); role ∈ {ADMIN, DOCTOR}

Implementation notes
• All tables have created_at TIMESTAMPTZ NOT NULL DEFAULT CURRENT_TIMESTAMP.
• SERIAL/BIGSERIAL primary keys are generated by PostgreSQL.
• prescriptions.prescribed_by and risk_assessments.performed_by are text fields, not FKs to user_accounts.
• assessment_findings severity/mechanism fields are point-in-time snapshots.
• Source ATC codes are stored as provenance text in source_atc_codes.
• Counters in risk_assessments are trigger-maintained on finding insertion; RBAC intends append-only clinician access.

Legend
PK = Primary Key (underlined)
FK = Foreign Key (arrow points to referenced table)
Fields marked snapshot are intentionally stored at assessment time.
Only database-enforced FKs are shown as connectors.